Fixed-size resultant roots and two Lean disproofs

Over an algebraically closed field, with positive declared sizes m, n and actual degree bounds $\deg f \leq m$, $\deg g \leq n$, Lean proves the following criterion. [pcc:unverified]

$$\mathrm{Res}_{m,n}(f,g) = 0 \iff (\exists a : \ f(a) = g(a) = 0) \ \vee \ (f_m = g_n = 0).$$

The sizes remain fixed when actual degrees drop. A vanishing resultant need not give a common finite root. The corrected criterion and both negative controls are fully quantified in the combined Lean binding. [pcc:unverified]

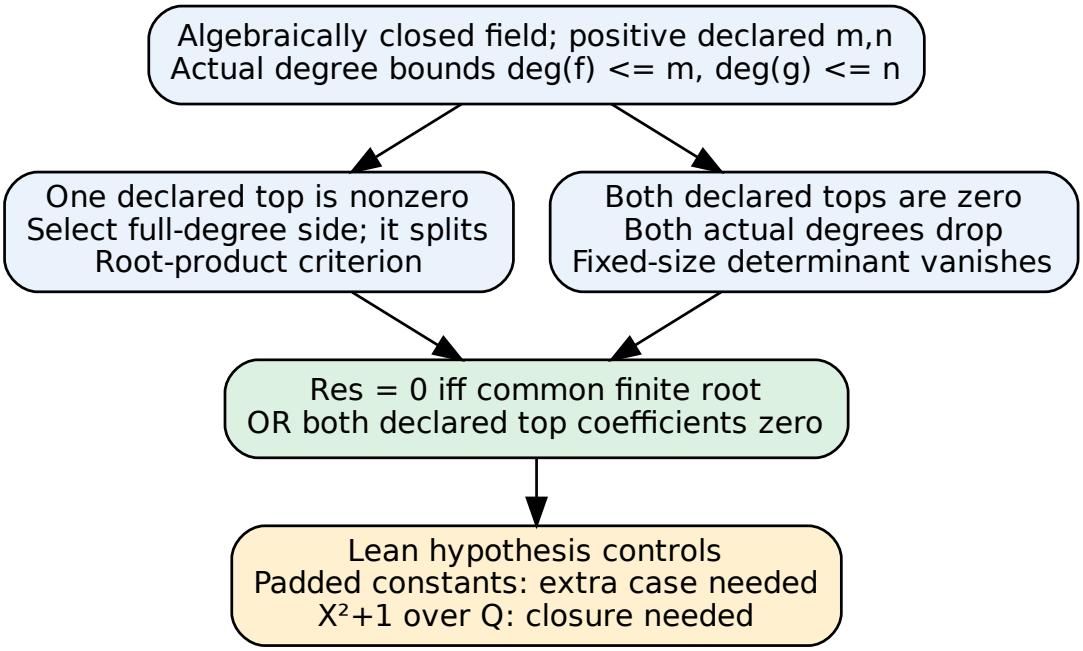


Figure 1: Actual algebraic case split and the checked domain controls. [pcc:unverified]

What the controls actually disprove

Exact Lean instance	Actual degrees	Declared sizes	Common finite root
$f = g = 1$, every field	0,0	2,2	None
$f = g = X^2 + 1$ over $\mathbb{Q}$	2,2	2,2	None

Both resultants are zero. The first has both declared top coefficients zero and directly refutes the universal affine-only criterion, even over algebraically closed fields. The second has leading coefficients 1,1 and shows why the converse needs splitting or algebraic closure. These are kernel proofs, not field samples. [pcc:unverified]

If one top coefficient is nonzero, that polynomial has its full declared degree. Over an algebraically closed field it splits. Mathlib's actual root-product formula expresses the resultant as a nonzero leading factor times the product of evaluations at its roots. If both tops vanish, both actual degrees are strictly smaller and the fixed-size Sylvester determinant vanishes. Resultant commutation handles the opposite full-degree side without assuming monicity or separability. [pcc:unverified]

Main binding: `declared_resultant_root_audit_certificate` in DeclaredResultantRootsAudit. The proof map is `../../DECLARED_ROOTS.md`; exact statement/source hashes are in `../../declared-root-sources.json`. Whole census gate431passes, zero failures/zero trusted-native exceptions. Local source commit `1029720b` precedes packet `PCC-20261110-2b8991`. Actual local verdict and unsigned standing are recorded separately. [pcc:unverified]

Lean 4.35.0 / Mathlib `56a00585` are pinned. Complete source 02 compiles with no warnings and only propext, Classical.choice and Quot.sound. Attempt 01's failed coefficient-one simplifications are retained with exact source, raw output and exit status. No hole, custom axiom, native decision exception or linter waiver. [pcc:unverified]

Sources reviewed: Mathlib Resultant.Basic root-product and degree-padding lemmas, IsAlgClosed.Basic splitting definition, Polynomial.Roots, and the existing actual GeneralS4 common-root/gluing and F5 constant-model counterexample. Prior Semaev primary read scopes are unchanged. Original Semaev (2004) remains unread after 403. No novelty or source-status promotion. [pcc:unverified]

This is the explicit finite-root/leading-zero polynomial theorem. It does not construct projective space or finish the actual elliptic point-list zero-sum recursion. Canonical evaluation and point-list induction remain the next bridges. No native experiment, timing or paid compute. [pcc:unverified]